



Service Procedure No.:	03-05
Service Procedure Title:	Laser Alignment
Applicable Systems:	8000, Micrion 9800 FlipChip, MicroMill HT
Frequency of Service:	As Needed

Introduction

Some FIB systems are equipped with Laser IF stage location systems. This procedure describes how to align the laser system so that it indicates stage motion accurately.

Materials Required

This procedure requires the use of the following materials and equipment:

- Standard field service engineer's tool set
- Clean-room gloves
- Lint-free wipes
- Laboratory-grade isopropyl alcohol
- Dial indicator, Federal Corp. LT-150 or equivalent, accurate to 0.00005" (1.28 μm)
- Adjustable dial indicator stand
- A second person to help with the alignment
- HP laser target(s)
- HP laser alignment aid tool

This procedure refers to Service Procedure 06-03, Workchamber Inspection and Cleaning, 03-01, Stage Travel Limit Switch Inspection, and 01-02, Power Down, Vent, and Reset.

NOTICE

Per paragraph 11.1 of Safety Guidelines for Semiconductor Manufacturing Equipment (S2-93), this procedure is a Type 3 task. Specifically:
"Equipment is energized. Live circuits are exposed and accidental contact is possible. Potential exposures are less than 30 volts RMS, 42.2 volts peak, 240 volt-amps, and 20 Joules."

Part 1. System Preparation

Configure the Chemical Sources in “Safe” Mode

Before performing the tasks described in this procedure, you must put the chemical sources in a “safe” mode; see Service Procedure 02-01.

WARNING

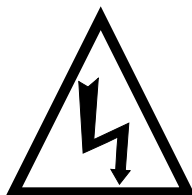


Do not proceed until the chemical sources are secure. The chemical reservoirs in FIB systems contain hazardous materials. Labels affixed to the exterior of a chemical cabinet identify the chemical(s) contained inside. If a chemical source is not in its “safest” state, as described in **Service Procedure 02-01**, it is possible for a combination of component failures to release the chemical(s) through a cabinet’s delivery line, a roughing line, a purging line, or from a chemical reservoir into the workchamber. Do not open the workchamber, or disassemble any part of the vacuum pumping system, or detach any gas line from any chemical cabinet until you are sure that every chemical source attached to the FIB system is in its “safest” state.

Power Down and Vent the System

Before performing the tasks described in this procedure, you must power down and vent the system; see Service Procedure 01-02.

WARNING



Dangerously high voltages are present in the process module while the FIB system is powered up. Do not rely *only* on the software Power Down function. Disable the voltages using both manual and automatic procedures, as described in **Service Procedure 01-02**. Do not proceed until you have switched off the applicable power supplies.

Open the Workchamber

Remove the front panels to gain access to the workchamber door.

When the system is completely vented, open the workchamber lid by removing the 1/2” socket head cap screws.

WEAR GLOVES, DO NOT HANDLE SOLVENTS

When working inside the chamber or when handling any components that will be exposed to vacuum, wear gloves rated for Class 100 environments or better. Due to disintegration of such gloves by solvents, do not handle solvents or parts that are wet with solvent.

CAUTION

The laser incorporated in the FIB System is a Class 2 (low power) laser. Eye damage can result if you look directly into the beam or into a specular (mirror-like) reflection of it. Do not stare into the beam or into a specular reflection of it.

Remove the Column

You must remove the column in order to gain access to the reference mirror adjustment screws^[1]; see service procedure 05-09-05 (for a 5 nm column), 05-09-10 (for a 10 nm column), or 05-09-25 (for a 25 nm column).

[1] If you are reasonably certain that the positions of the reference mirrors have not been altered, it may not be necessary to remove the column and adjust the reference mirrors in order to perform a successful alignment.

Part 2. Stage Mirror Alignment

Note: An indication of stage mirror alignment can be obtained by using a multimeter to check the voltage at the pins on the back of the receiver(s) while moving the stage to its positive and negative limits (in x and y). A lack of change in output voltage indicates that the stage mirrors are fairly well aligned.

- 2-1. Using the *motorshell* program, move the stage to its positive x limit. Service Procedure 03-01, Stage Travel Limit Switch Inspection, describes how to use the *motorshell* program to move the stage. It is also possible to move the stage by turning the ends of the x and y drive motor shafts (outside of the workchamber) by hand or with a 3/8" box wrench, but it is recommended that you use the *motorshell* program to move the stage.
- 2-2. Figure 1 shows the position of the x-axis stage mirror within the workchamber. Figure 2 shows the side view of the x-axis stage mirror. Note that there is a scribe line near the bottom of the mirror. The laser beams make contact with the area of the mirror above the scribe line; **this surface must not be scratched or damaged in any way.** The area below the scribe line is not contacted by the laser beams; this is the area of the mirror that will be used (with the dial indicator) to check the stage mirror alignment.
- 2-3. Use the adjustable dial indicator stand to position the dial indicator near the side of the x-axis stage mirror (see Figure 1). Adjust the height of the dial indicator stand so that the dial indicator will make contact with the area of the x-axis stage mirror **below the scribe line.** See Figure 2.

NOTICE: KEEP THE DIAL INDICATOR BELOW THE SCRIBE LINE

The area of the stage mirror above the scribe line must not be scratched or damaged in any way. Avoid contacting this area; keep the dial indicator below the scribe line.

- 2-4. Carefully move the dial indicator against the x-axis stage mirror until its dial begins to move. "Zero" the dial indicator by moving it so that its dial is aligned with one of the divisions (lines) on the dial indicator's gauge.
- 2-5. Move the stage back and forth in the y direction (using the *motorshell* program, or by turning the end of the y-axis drive motor shaft) while observing the dial indicator's gauge. The dial indicator reading should not vary by more than 0.00005" (1.28 μm) over the entire length (at the level of the scribe line) of the x-axis stage mirror.
- 2-6. If the dial indicator reading varies by more than 0.00005", use the stage mirror adjustment screws (shown in Figure 2) to adjust the position of the x-axis stage mirror. Begin with minimal rotation (less than one turn) of the stage mirror adjustment screws; even a small adjustment may result in a significant change in stage mirror alignment. Repeat step 2-5. Repeat the x-axis stage adjustment until the dial indicator reading is within specification.
- 2-7. Repeat the alignment procedure for the y-axis stage mirror.

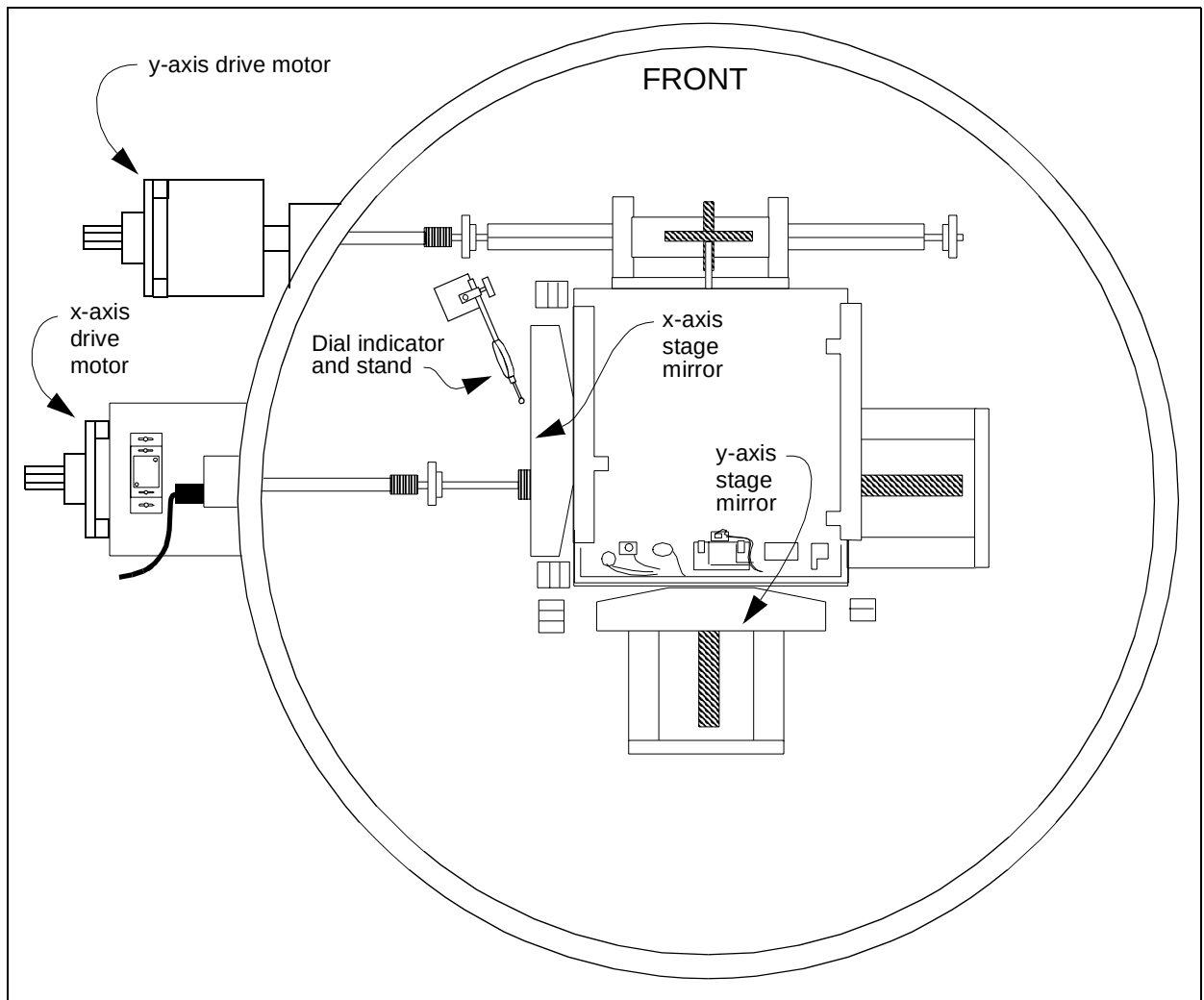


Figure 1. Top View of the Workchamber and Stage Mirrors (Lid Open)

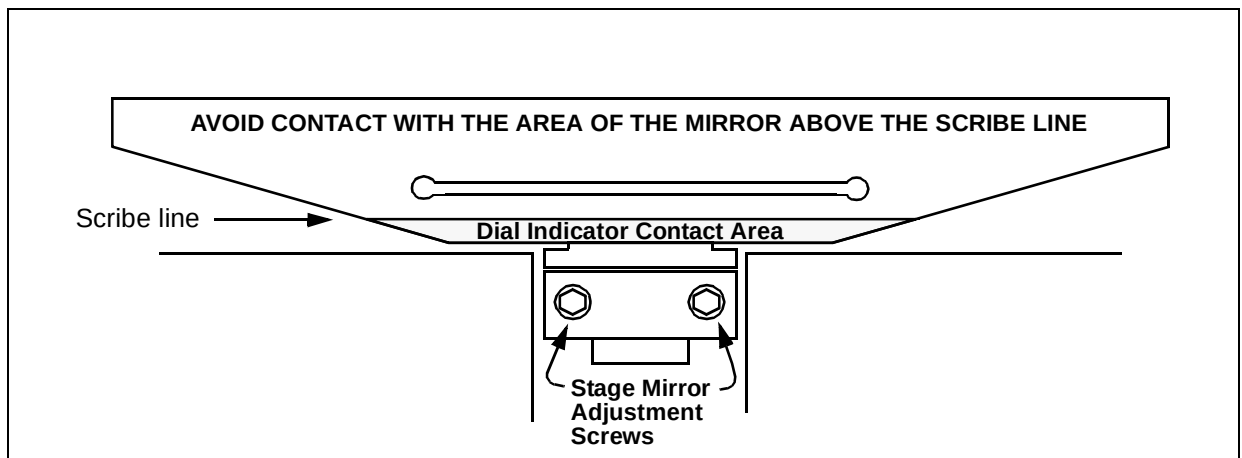


Figure 2. Side View of Stage Mirror

Part 3. Interferometer Alignment

Each interferometer should be flush against the workchamber and centered within its mounting slot (see Figure 3).

- 3-1.** Inspect each interferometer (see Figure 4 for the location of the x and y interferometers).
- 3-2.** If an interferometer is not properly aligned, loosen its mounting screws, adjust the position of the interferometer so that it is flush against the workchamber and centered within its mounting slot, then retighten the screws (see Figure 3).

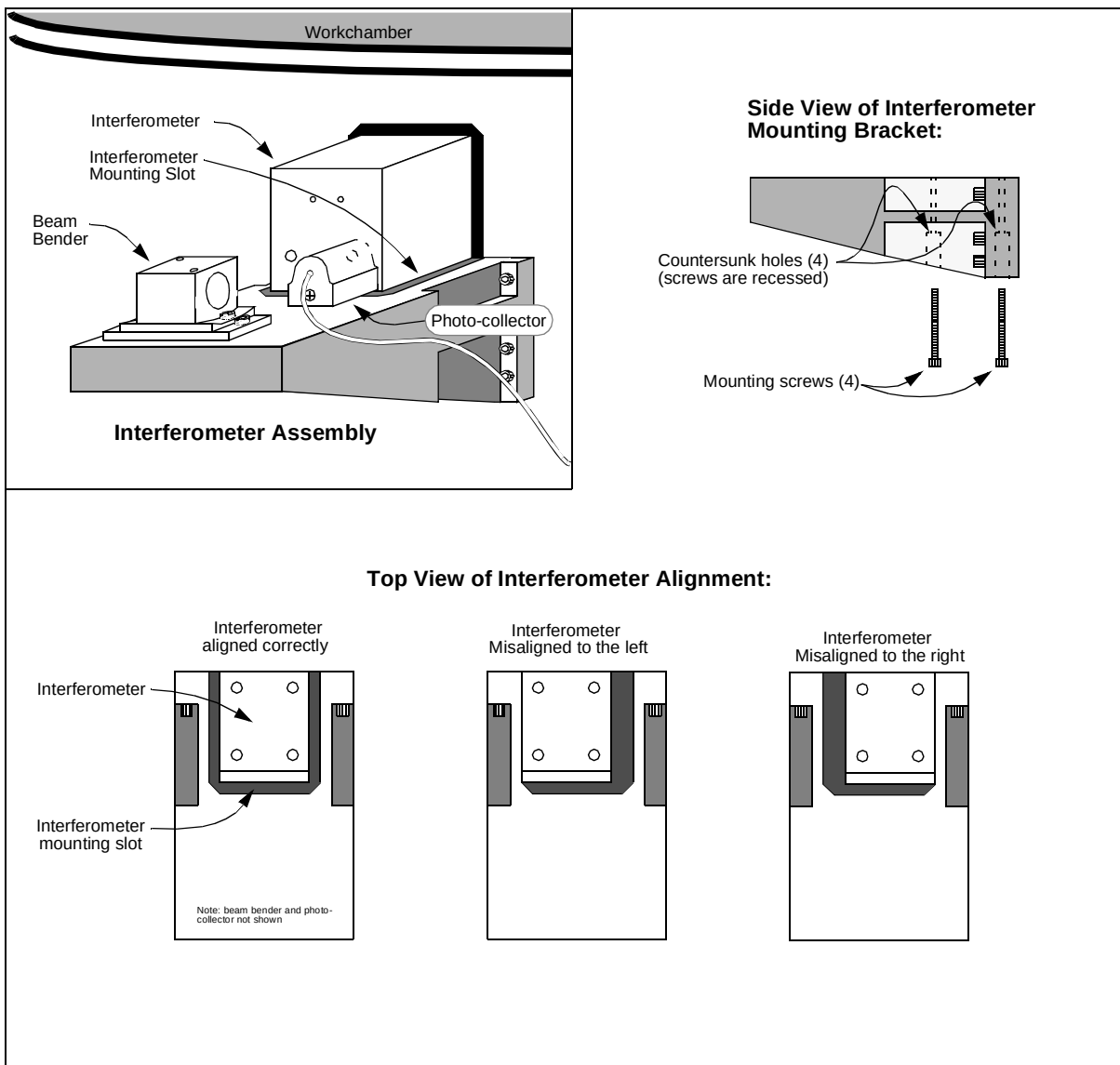


Figure 3. Top and Side Views of Interferometer

Part 4. Laser Alignment

Close the Workchamber

The workchamber must be closed in order to perform the laser alignment procedure; close the workchamber lid and tighten the “T” screw on the front of the lid (so that the switch block on the lid actuates the switch on the top right of the workchamber).

- 4-1.** Using the *motorshell* program, move the stage to the left front corner of the workchamber (as viewed from the front of the workchamber). Service Procedure 03-01, Stage Travel Limit Switch Inspection, describes how to use the *motorshell* program to move the stage. It is also possible to move the stage by turning the ends of the x and y drive motor shafts (outside of the workchamber) by hand or with a 3/8” box wrench, but it is recommended that you use the *motorshell* program to move the stage. See Figure 4 for the location of the laser system components.

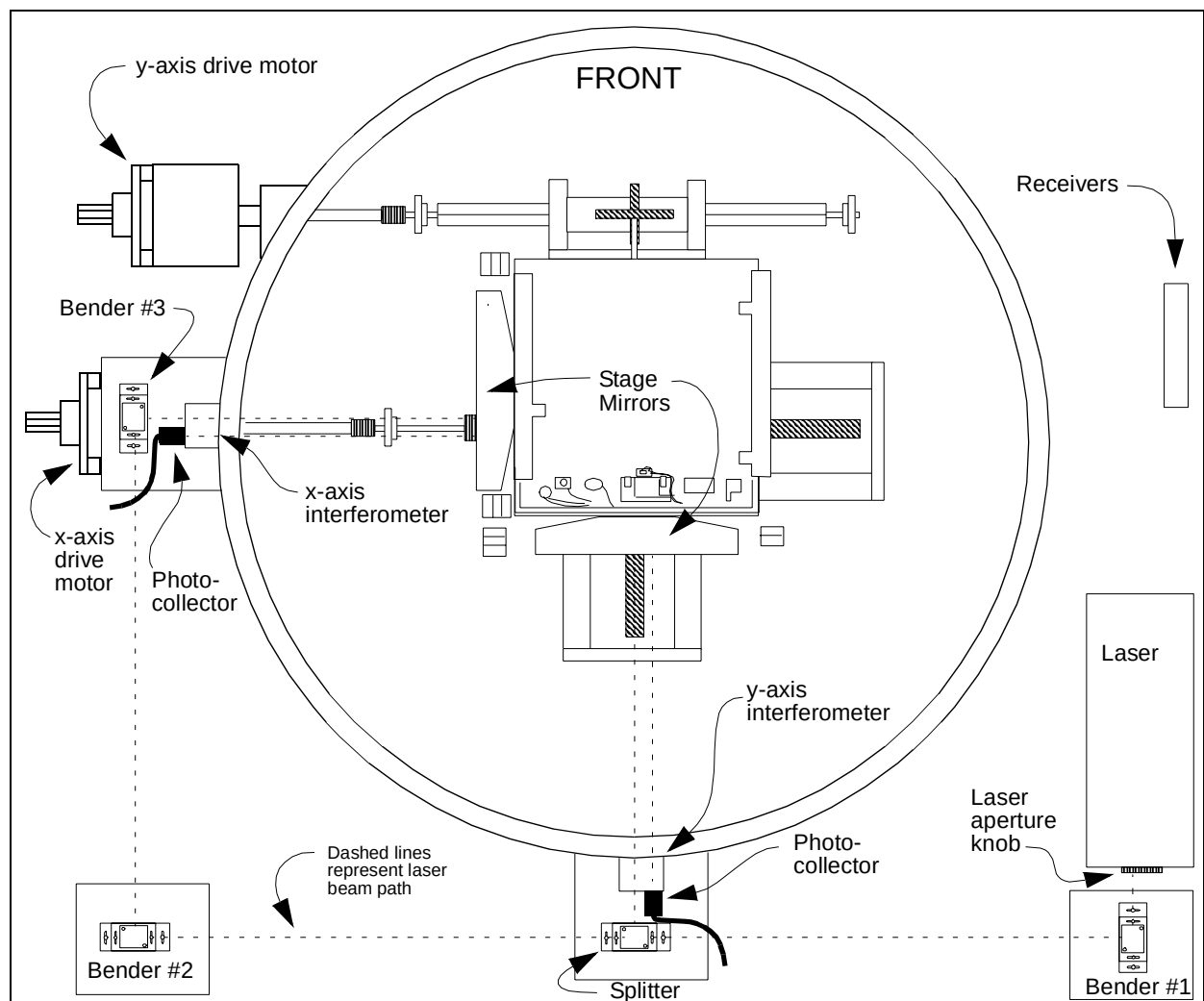


Figure 4. Top View of the Workchamber and Laser IF System (Lid Open)

- 4-2.** Turn on the laser system power.

4-3. Remove the laser tube shrouds and interferometer covers.

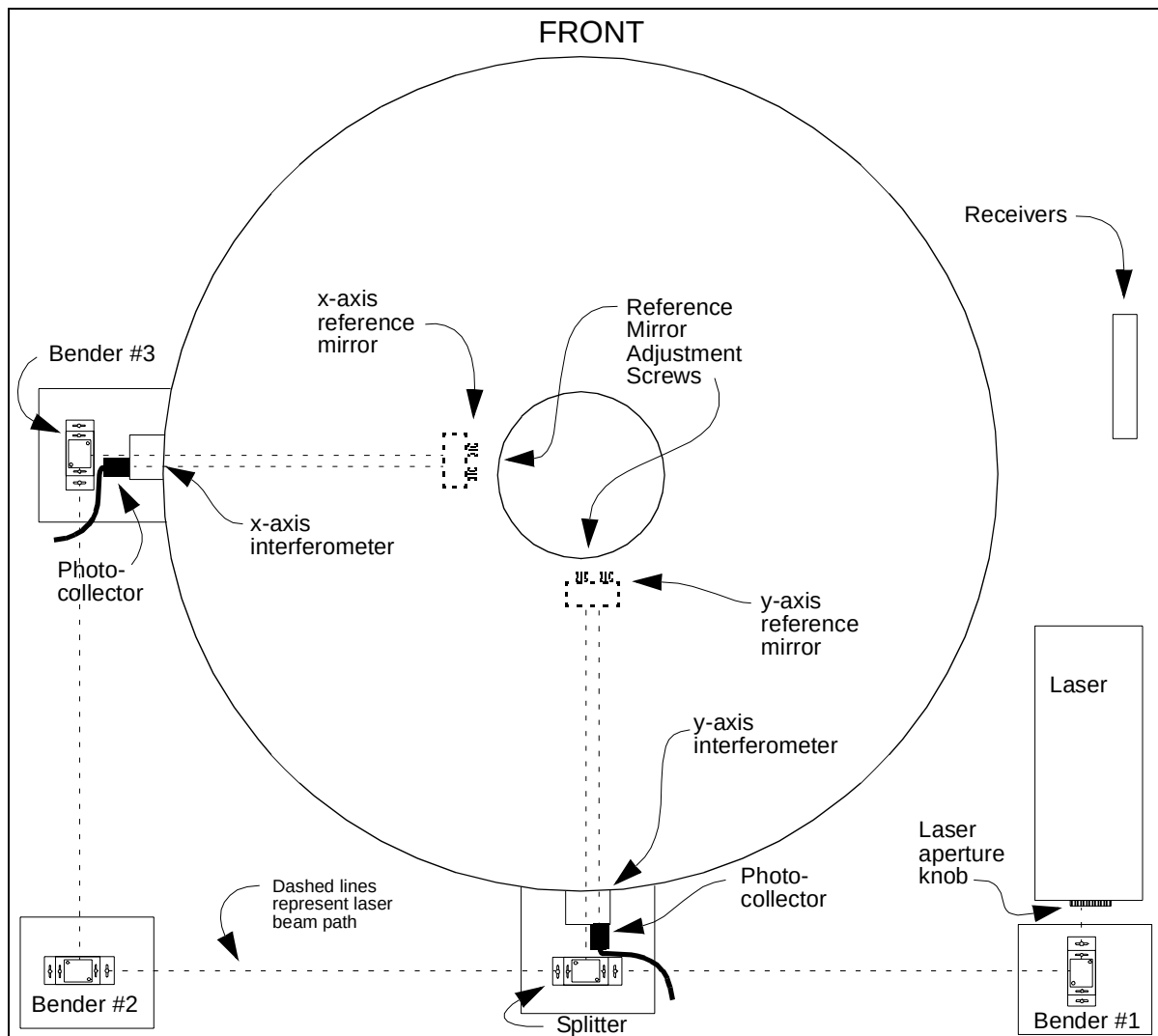


Figure 5. Top View of the Workchamber and Laser IF System (Lid Closed)

- 4-4.** Use a small slotted screwdriver to rotate the potentiometers on the sides of the receivers fully clockwise (to maximize gain).
- 4-5.** Use a multimeter to check the voltage at the pins on the back of the receiver. An output voltage of 1.5 V (minimum) is an indication that the laser system is fairly well aligned, but the system may still require alignment.
- 4-6.** Rotate the aperture knob on the laser to the small aperture position. Hold a piece of plastic or paper in front of the beam. Rotate the aperture knob (slightly) until the spot on the paper or plastic is as bright and as round as possible. *Do not rely on the line(s) on the aperture knob for output alignment; they should only be used as a rough indication of aperture position.*
- 4-7.** Hold an HP laser target over the input to beam bender #1. Loosen the laser mounting bolts

and adjust the laser position until the beam is centered on the laser target (and therefore the input to beam bender #1). If necessary, loosen the two socket head screws on bender #1's mounting plate to adjust the lateral position of bender #1 (so that the beam is centered on the input to bender #1), then tighten them again. Figure 6 provides detail on the location of the beam bender lateral and rotational adjustment screws. Tighten the laser mounting bolts. Check the beam alignment again to make sure that nothing has moved.

- 4-8.** Adjust beam bender #1 (using a laser target) so that the beam output from bender #1 is centered on the input to bender #2 (Figure 4 shows the location of the beam benders). It may be necessary to adjust the rotational and lateral position of bender #1 in order to center the beam on the input to bender #2. If this is the case, check the beam alignment at the input to bender #1, and adjust if necessary.

Note: the goal is to get the beam centered on each bender without changing the alignment of previously aligned benders. Previous alignments are less likely to be affected if you only have to adjust the beam bender's rotational position.

- 4-9.** Adjust beam bender #2 (using a laser target) so that the beam output from bender #2 is centered on the input to bender #3. It may be necessary to adjust the rotational and lateral position of bender #2 in order to center the beam on the input to bender #3. If this is the case, check the beam alignment at the input to bender #2, and adjust if necessary.
- 4-10.** Adjust beam bender #3 so that the output of bender #3 is centered on the reference port of the x-axis interferometer (see Figure 6).
- 4-11.** Use a small Phillips-head screwdriver to loosen the x-axis photo-collector mounting screw. Remove the photo-collector from the measurement port of the x-axis interferometer.
- 4-12.** Place a piece of white paper or Scotch tape over the x-axis interferometer measurement port (see Figure 6). Adjust beam bender #3 so that the beam reflected off the stage mirror is centered on the measurement port and is as bright as possible. If no red glow is visible at the measurement port, it may be necessary to adjust the x-axis reference mirror (see Figure 5).
- 4-13.** Adjust beam bender #3 and the x-axis reference mirror so that the beam is reflected back into the measurement port, then do a more precise adjustment by maximizing the brightness of the red glow at the measurement port. Another technique is to slide a strip of glossy thermal printer paper into the slit in the side of the interferometer so that it covers the reference beam: this will diffuse the beam. Scotch tape must be used at the measurement port to detect the reflected glow (paper will obscure it). Then adjust the position of bender #3 and the x-axis reference mirror to maximize the brightness of the red glow at the measurement port.

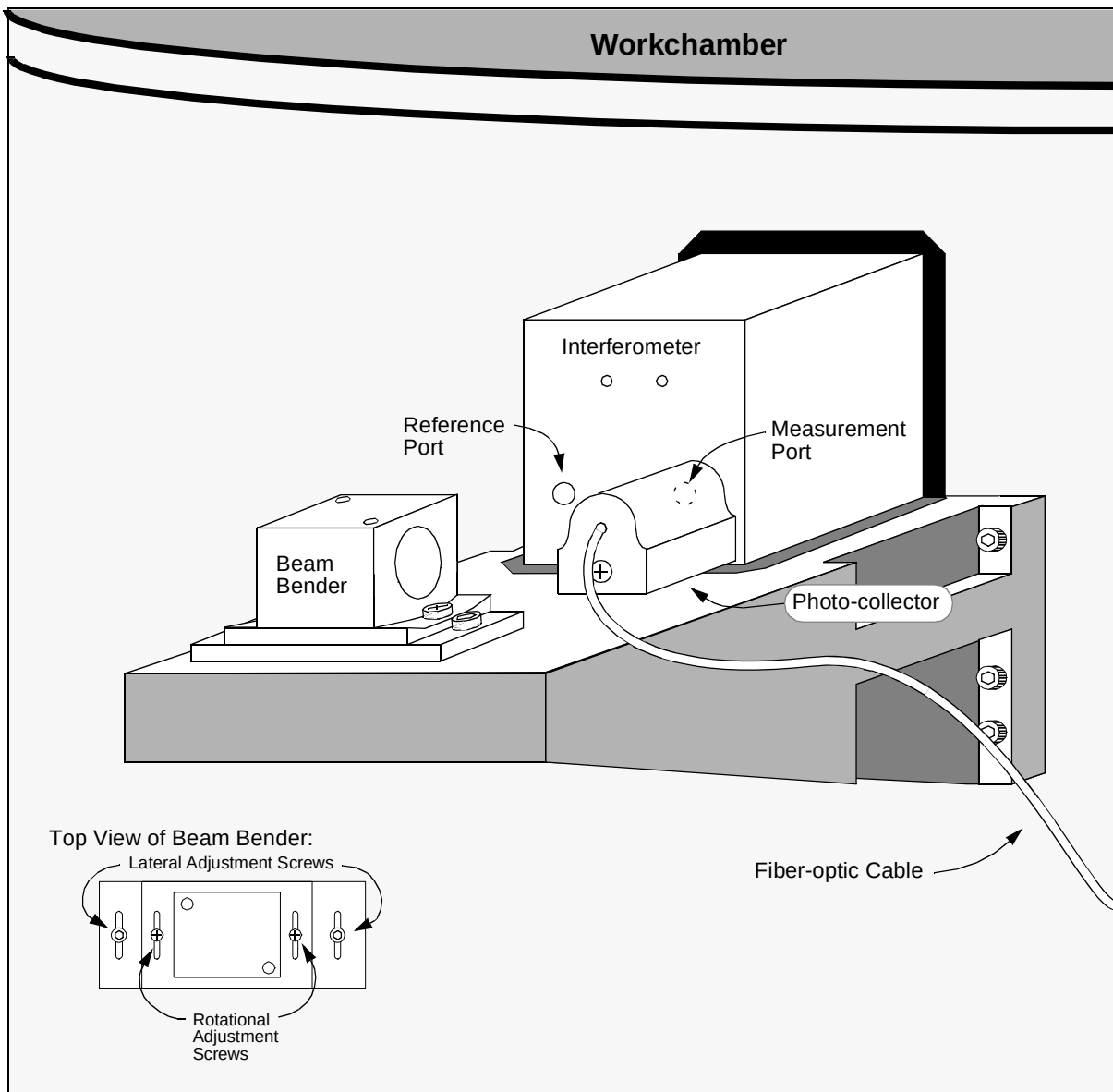


Figure 6. Interferometer, Photo-collector, and Beam Bender

- 4-14.** When the measurement output (glow) has been optimally adjusted, put the photo-collector back in place at the measurement port. Check the green LED on the side of the x-axis receiver; it should be illuminated. If the LED is not on, try to align bender #3 and the x-axis reference mirror again (repeat steps 3-9 through 3-12). If the LED is on, use a multi-meter to check the voltage at the pins on the back of the receiver; the output voltage should be 1.5 V (minimum). If the output voltage is not 1.5 V (minimum), try to align bender #3 and the x-axis reference mirror again.

Note: wires, cables, fingers, etc. can block the beam pathway. If you lose the signal completely, check for beam path obstructions before starting realignment. It is also possible to lose the signal if the interferometer fiber-optic cable is pinched (by a cable tie, etc.)

- 4-15.** When bender #3 is aligned so that the green LED on the side of the x-axis receiver is on and the receiver output voltage is 1.5 V (minimum), check to see that the beam is reflecting back to the laser. Insert an alignment aid tool (HP 10706-60202) into the slit in the side of the x-axis interferometer so that the clear plastic end of the tool covers the (output) of the reference beam. This should produce an “autoreflexion” of the beam.

CAUTION



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- 4-16.** Look **indirectly** into the output of the laser. A **very faint** (almost transparent) disk should be visible within the laser output port. If the disk isn't visible, move the (plastic end of the) alignment aid tool so that it alternately covers and uncovers the reference beam output: the disappearance and reappearance of the autoreflexion disk can aid in its detection.
- 4-17.** Repeat the alignment procedure for the y-axis.
- 4-18.** After the x-axis and y-axis beams have been aligned, rotate the laser output aperture to the large aperture. Ensure that the beam output is centered within the large aperture (the sides of the aperture opening should not interfere with the beam).
- 4-19.** Connect a multimeter to the pin on the back of the x-axis receiver. Use a small slotted screwdriver to rotate the potentiometer on the side of the receiver counterclockwise until the output voltage is approximately 1.0 V. Repeat for the y-axis receiver.
- 4-20.** Install the laser tube shrouds and interferometer covers.

